



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2025-26

Class:	BE	Semester:	VIII
Course Code:		Course Name:	AAI

Name of Student:	BARI ANKIT VINOD
Roll No. :	61
Assignment No.:	4
Title of Assignment:	
Date of Submission:	
Date of Correction:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Completeness	5	03
Demonstrated Knowledge	3	02
Legibility	2	02
Total	10	07

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Completeness	5	3-4	1-2
Demonstrated Knowledge	3	2	1
Legibility	2	1	0

Checked by

Name of Faculty : Dr. Vaishali Shinde

Signature :

Date :

[Signature]
6/4/24

Q. 1) A company has a small medical images dataset and wants to use a pre-trained CNN model by freezing all convolutional layers and training only the final classifier layers. Identify which transfer learning technique should be used and explain how it works?

→ The appropriate technique for feature extraction using a pre-trained CNN.

• Expⁿ -

In this approach, a CNN model has already been trained on a large dataset (like ImageNet) is reused. Since the company has a small medical image dataset, training a deep network from scratch would likely lead to overfitting and poor generalization.

• How It works -

- The convolution layers are frozen, means their weights are not updated during training.
- These acts as feature extractors, capturing low-level and high level (shapes, patterns) features.
- only the final classifier layers is trained on the new dataset.
- The extracted features are passed to this classifier, which learns to map them to the target medical classes.

• why this works -

- Pre-trained CNNs have already learned general visual features that transfer well to new tasks.
- freezing layers reduces the number of trainable parameters, less data required.
- Helps to avoid overfitting while improving performance on small dataset.
- Pre-trained CNNs have already learned general visual features that transfer well to new tasks.
- freezing layers reduces the number of trainable parameters, less data required for that.

Q. 2) An AI system is required to generate realistic synthetic images to augment a limited dataset for training a classifier. Identify which generative model should be applied and explain how its architecture helps solve the problem.

→ The most suitable model is a generative Adversarial network (GAN).

• Expⁿ.

• A GAN is a specifically designed to generate realistic synthetic data, making it ideal for data augmentation when dataset are limited.

• Architecture and how it helps -

• A GAN consist for two neural networks.

① Generator -

• Takes random noise as input.

• generates synthetic images resembling real data.

② Discriminator -

• Acts as a classifier.

• Tries to distinguish between real images and generated images.

• Training process -

• The generator tries to fool the discriminator by producing realistic images.

• The discriminator tries to correctly classify real vs fake images.

• Improves classifier performance by providing diverse training samples.

• Reduces overfitting caused by limited data.

• The generator generates images to fool the discriminator by generating realistic images.

• The discriminator tries to identify real and fake images, and both neural networks are improved.

Q7 Explain transfer learning, describe different types of transfer learning.

→ Transfer learning is a technique in machine learning where model is trained on one task is reused for different but related task.

Instead of starting from scratch, the model leverages previously learned knowledge, saving time, data and computational resources.

- why transfer learning is important?
- Reduces needs for large labeled dataset.
- speeds up training
- Improves performance when data is scarce
- enables knowledge reuse across domains.

- Types of transfer learning.

(i) feature extraction -

- Pre-trained model is used as a fixed feature extractor.
- only the final layers are trained.
- Eg. freezing CNN layers, and training classifiers.

(ii) fine-tuning -

- Some or all pre-trained layers are unfrozen and retrained.
- Allows the model to adapt more specifically to the new task.
- Used when the new dataset is larger or similar to the original dataset.

(iii) Domain adaptation -

- Used when the source and target domains differ (eg. natural images vs medical images)
- Model is adapted to handle domain shifts.

(iv) Task transfer -

- Source and target tasks are different but related.
- Eg. using an image classification model for object detection.

(v) Unsupervised transfer learning -

- knowledge transfer from unlabeled data.
- often used in representation learning.
- Transfer learning enables models to reuse knowledge from large database and apply it effectively to new problems, especially when data is limited. Techniques like feature extraction and fine-tuning are widely used in real world AI applications.

- fine-tuning -

- fine-tuning a large language model (LLM) is the process of retraining a pre-trained model (or a foundation model) for a specific task, dataset, tone or format.

(i) RAG - easy to update knowledge without retraining,

- can handle large and dynamic dataset.
- lower cost compared to fine-tuning.
- slower at runtime.
- output depends on quality of retrieval.
- may give inconsistent style/format.

(ii) fine-tuning - produces consistent styles and responses,

- faster at runtime.
- tailored for very specific task.
- expensive and time-consuming to retrain.
- hard to update.